Avantika Varma

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Education

University of California, Davis Davis, CA
Bachelor of Science in Computer Engineering, Minor in Business 2023-2027

Experience

Solidigm - SK Hynix Rancho Cordova, CA
NAND Design Automation Engineering Intern March 2026 – Present

- Architected an end-to-end RTL verification EDA flow in Cadence Xcelium, built on Python-driven testbench automation, automated stimulus generation, coverage collection, and regression execution scripts that reduced manual verification setup time by 40%.
- Developed a Python and Bash automation framework on Linux for parallel regression testing across multiple NAND device configurations; optimized the execution pipeline to achieve 6x faster test cycles through Python-based job scheduling, batch processing, and automated failure detection.
- Built an agentic AI-assisted debugging pipeline in Python with case-specific agents, reducing debug turnaround time by 65% while improving failure classification accuracy.
- Converted 5+ legacy Perl verification scripts into modern Python architecture; designed Python data processing pipelines for Dataiku enabling real-time monitoring of regression health across all design teams.
- Managed automation code in Git and GitHub, and coordinated design collaboration across verification, design, and physical design teams.

California-Pacific-Northwest AI Hardware Hub Davis, CA
Research Intern, Integrated Nano Systems Lab April 2025 – March 2026

- Optimized photonic grating structures using Python-scripted Ansys Lumerical FDTD simulations; evaluated 50+ geometry variants through automated parameter sweeping and statistical analysis to maximize light concentration
- Designed bio-inspired 3D photonic geometries CAD models and developed Python conversion pipeline for 3D model generation

Tripathi Physics Lab (UC Davis) Davis, CA
Student Researcher September 2024 – June 2025

- Engineered Python automation framework (PyVISA, socket programming, NumPy) to autonomously control 16-channel CAEN digitizers, waveform generators, and power supplies via Ethernet; reduced manual hardware setup time by 50% and improved measurement repeatability to 98%.
- Developed pulse shape discrimination (PSD) algorithms in Python using timing-based feature extraction and statistical analysis to differentiate neutron vs. gamma event signatures with 92% accuracy across 10K+ waveforms
- Debugged hardware interfaces using oscilloscopes and multimeters, achieving stable signal integrity.

The Make Box (B2C STEM Startup) Davis, CA
Product Innovation & Engineering Outreach Lead January 2024 – Present

- Conducted 50+ customer discovery interviews with educators and students; synthesized qualitative feedback into structured product requirements, shaped MVP development roadmap based on validated customer needs.
- Developed and iterated MVP across 150+ early adopters and 3 product variations; implemented user-driven feature changes through rapid prototyping and A/B testing analysis
- Secured $12,500 in funding and Big Bang! Business Competition win by presenting customer pain point analysis, product-market fit metrics, and go-to-market strategy.

University of California Davis College of Engineering Davis, CA
Peer Advisor March 2026 – Present

- Mentored 20+ undergraduate engineering students on degree planning and course sequencing; translated complex academic policies into actionable guidance and technical course recommendations.

Projects

AI-Powered Personal Planner Tool July 2026 – Present
Claude AI, HTML, JSON, Google Calendar API, GitHub Pages

- Leveraged Claude to rapidly prototype and iterate on a custom productivity tool tailored to personal workflows, accelerating development by focusing on feature logic rather than boilerplate implementation.

- Designed and built a responsive web application with dynamic calendar views, integrated to-do lists, and bidirectional Google Calendar synchronization to centralize task and event management.
- Created a personalized user interface using modern web design principles, demonstrating how AI-assisted development enables quick validation of product ideas and rapid customization based on user needs.

Real-Time Stock Ticker Game April 2026 – June 2026

C++, Firmware, Microcontroller, I2C, SPI, AWS IoT, APIs, LED Display

- Designed and programmed embedded firmware in C++ for a microcontroller-based stock ticker that delivered real-time market-style updates to an LED display.
- Implemented I2C and SPI communication protocols to interface reliably with components and drive display.
- Integrated external APIs and AWS IoT services to retrieve, transmit, and process live data streams, enabling responsive game behavior and cloud-connected functionality.

Die Counting Game on FPGA March 2025 – June 2025

Verilog HDL, Intel Quartus Prime, ModelSim, FPGA Hardware

- Designed and implemented a digital dice game on an Altera MAX FPGA using Verilog, with modular components for pseudo-random number generation and finite state machine (FSM) control logic for custom seven-segment displays.
- Applied verification methodologies in ModelSim by building test benches to validate logic correctness, timing behavior, and edge cases.
- Performed physical testing on FPGA hardware to confirm signal integrity and verify that the synthesized logic was correctly mapped to device resources.

AI-Powered Data Analytics Dashboard April 2025

Streamlit, Python, Gemini AI, Pandas, NumPy

- Developed a modular business intelligence platform in Streamlit and Python to transform raw nonprofit datasets into actionable insights through automated visualization.
- Integrated Gemini AI via API to support natural language queries, enabling users to generate summaries, visuals, and insights directly from their data.

Music-Following Robot January 2025 – March 2025

TI MSP430, C++, Code Composer Studio, Analog Electronics

- Designed an autonomous robot using an MSP430 microcontroller and implemented C++ firmware for real-time noise sampling and pulse width modulation (PWM) motor control.
- Built active filtering stages with operational amplifiers to isolate target musical frequencies and guide motion.
- Diagnosed and resolved circuit issues by testing continuity, voltage levels, and current flow across electronic components.

Awards & Honors

- **Big Bang! Business Competition Winner** (2025) – $12,500 in funding and startup residency awarded
- **Best Project Award** (2025) – California-Pacific-Northwest AI Hardware Hub and CITRIS showcase
- **Dean's Honors List** (Fall 2023) – Awarded to the top 16% of the college

Skills

Languages: Python (NumPy, pandas, PyTorch, scikit-learn, PyVISA), C++, Verilog, SystemVerilog, Perl, Bash, MATLAB, C
EDA & Tools: Cadence (Xcelium, Verdi), Synopsys, Design Sync, Intel Quartus, ModelSim, Dataiku, Git, Linux/Unix, Blender, Ansys Lumerical
Specializations: RTL verification flows, design automation, Python automation frameworks, regression testing, LVS/DRC scripting, agentic AI workflows, hardware automation, signal processing, real-time data pipelines